Templates for 3D Object Pose Estimation Revisited: Generalization to New Objects and Robustness to Occlusions

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https://nv-nguyen.github.io/template-pose

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Pytorch code

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Motivation

Goal → estimate 3D pose of new objects with only their 3D models

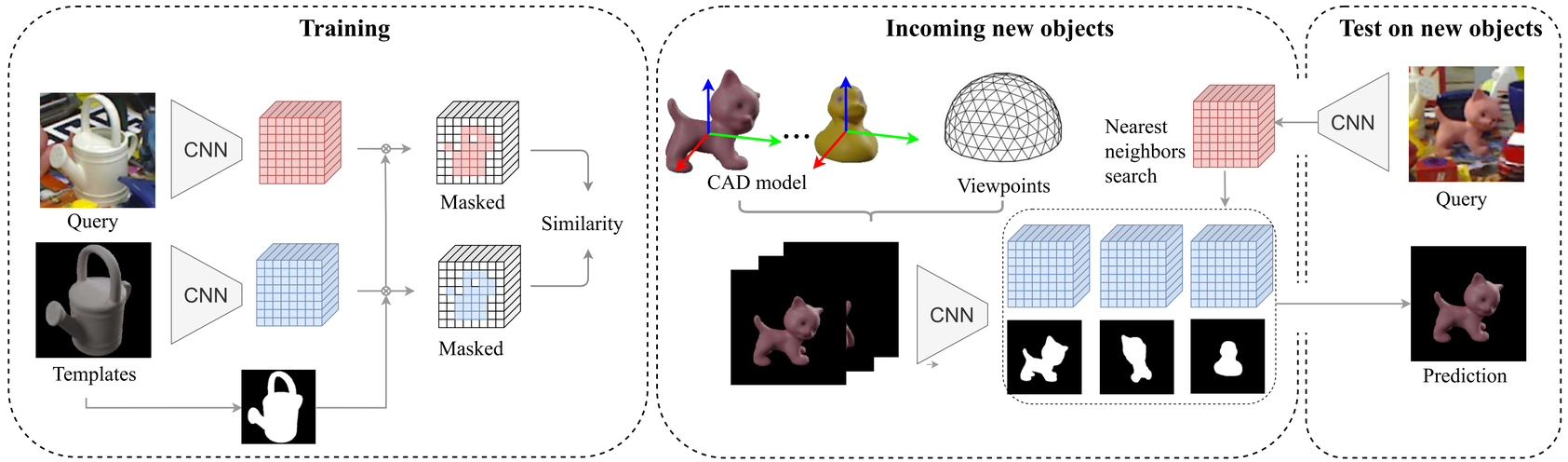
Previous works → require training phase or similar to training objects

1. Known object pose estimation: Wohlhart and Lepetit [1], Balnas et al. [2], BB8 [5], CosyPose [6], etc;
2. Category level pose estimation: StarMap [5], NOCS [6], etc;
3. Pose estimation on T-LESS: Sundermeyer et al. [7, 8].

Our contribution

1. Failure-case analysis of previous template-based methods when testing on new objects;
2. A method that can predict the pose of new objects from their CAD models, without training on these objects nor restricting these objects to be similar to the training ones;
3. A method robust to occlusions even in the challenging scenario when objects are both new and occluded.

Method



Local feature similarity

Training

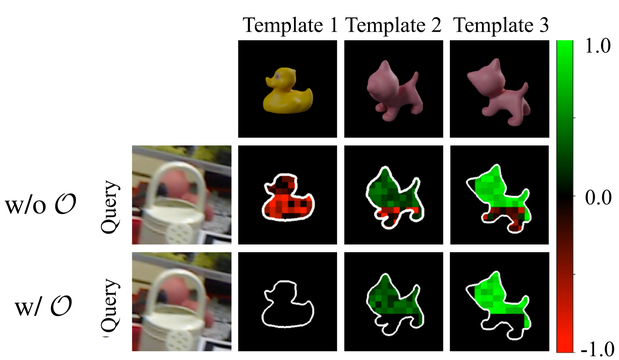
$$\mathrm{sim}(\bar{\mathbf{q}}, \bar{\mathbf{t}}) = \frac{1}{|\mathcal{M}|} \sum_l \mathcal{M}^{(l)} \mathcal{S}\left(\bar{\mathbf{q}}^{(l)}, \bar{\mathbf{t}}^{(l)}\right)$$

$$\text{where} \quad \mathcal{S}\left(\bar{\mathbf{q}}^{(l)}, \bar{\mathbf{t}}^{(l)}\right) = \frac{\bar{\mathbf{q}}^{(l)}}{||\bar{\mathbf{q}}^{(l)}||} \cdot \frac{\bar{\mathbf{t}}^{(l)}}{||\bar{\mathbf{t}}^{(l)}||}$$

Run-time

$$\mathrm{sim}^*(\bar{\mathbf{q}}, \bar{\mathbf{t}}) = \frac{1}{|\mathcal{M}|} \sum_l \mathcal{M}^{(l)} \mathcal{O}^{(l)} \mathcal{S}\left(\bar{\mathbf{q}}^{(l)}, \bar{\mathbf{t}}^{(l)}\right)$$

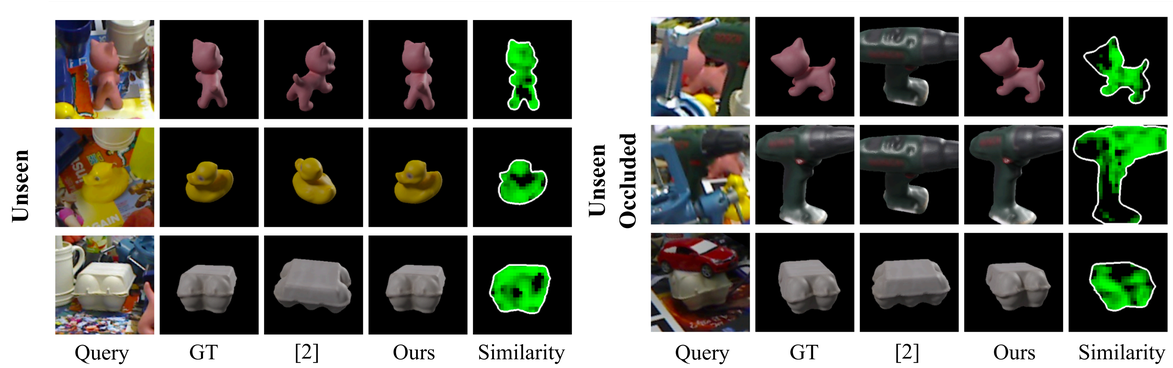
$$\text{where} \quad \mathcal{O}^{(l)} = 1_{\mathcal{S}(\bar{\mathbf{q}}^{(l)}, \bar{\mathbf{t}}^{(l)}) > \delta}$$



Results

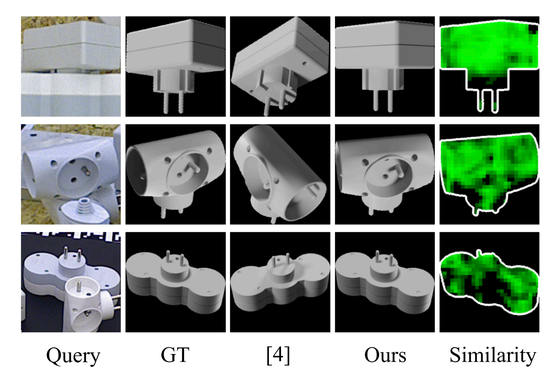
LINEMOD (LM) and Occlusion-LINEMOD (O-LM)

Method	Backbone	Features	Loss	Seen LM				Seen O-LM				Unseen LM				Unseen O-LM			
				#1	#2	#3	Avg.	#1	#2	#3	Avg.	#1	#2	#3	Avg.	#1	#2	#3	Avg.
[1]	Base [1]	Global	[1]	87.0	83.1	85.1	85.0	19.2	23.1	15.0	19.1	13.2	15.5	18.2	15.2	9.3	5.1	5.1	6.5
[1]	Base [1]	Global	InfoNCE	95.2	95.3	95.4	95.3	19.6	25.3	16.1	20.3	13.3	17.0	20.5	16.9	8.2	6.4	6.7	7.1
[2]	Base [1]	Global	[2]	89.2	85.4	83.3	86.3	18.3	21.9	17.6	19.5	14.1	16.3	19.7	16.7	8.2	7.5	7.6	7.8
[2]	Base [1]	Global	InfoNCE	96.3	95.2	96.5	96.0	18.3	23.1	15.8	19.1	11.5	17.7	17.2	15.5	7.1	6.5	6.5	6.7
Ours	Base [1]	Local	[1]	84.8	85.5	86.3	85.5	50.1	51.3	42.2	47.9	69.6	63.2	46.2	59.7	35.3	34.3	44.2	37.9
Ours	Base [1]	Local	InfoNCE	95.6	96.9	92.0	94.8	68.9	71.0	57.7	65.8	78.8	82.5	64.1	75.1	42.2	57.1	59.8	53.0
[1]	ResNet50	Global	InfoNCE	98.8	96.9	98.8	98.1	66.7	73.2	62.7	67.5	42.2	43.7	49.4	45.1	22.3	22.5	45.9	29.9
[2]	ResNet50	Global	InfoNCE	96.9	97.1	94.5	96.1	63.6	71.8	58.9	64.7	39.9	44.9	48.3	44.3	15.5	21.8	50.2	29.1
Ours	ResNet50	Local	InfoNCE	**99.3**	**99.0**	**99.2**	**99.1**	**77.3**	**84.1**	**76.8**	**79.4**	**94.4**	**97.4**	**88.7**	**93.5**	**71.4**	**72.7**	**85.3**	**76.3**



T-LESS

Method	Number templates	Recall VSD		
		Obj. 1-18	Obj. 19-30	Avg
Implicit [3]	92K	35.60	42.45	38.34
MPL [4]	92K	35.25	33.17	34.42
Ours	92K	**59.62**	**57.75**	**58.87**
Ours	21K	59.14	56.91	58.25



[1] Learning Descriptors for Object Recognition and 3D Pose Estimation, Wohlhart and Lepetit, CVPR 2015
[2] Pose Guided RGBD Feature Learning for 3D Object Pose Estimation, Balnas et al., ICCV 2017
[3] Implicit 3D Orientation Learning for 6D Object Detection from RGB Images, Sundermeyer et al., ECCV 2018
[4] Multi-path Learning for Object Pose Estimation Across Domains, Sundermeyer et al., CVPR 2019
[5] BB8: a scalable, accurate, robust to partial occlusion method for predicting, Rad et al., ICCV 2017
[6] CosyPose: Consistent multi-view multi-object 6D pose estimation, Labbé et al., ECCV 2020
[7] StarMap for Category-Agnostic Keypoint and Viewpoint Estimation, Zhou et al., ECCV 2018
[8] NOCS for Category-Level 6D Object Pose and Size Estimation, Wang et al., CVPR 2020